

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

Jnana Sangama, Belagavi - 590 018, Karnataka



AI & Full Stack Development Intern

A Report submitted in partial fulfillment of the requirements for the Course

Internship (Course Code: 24AM8SRINT)

In the Department of
Machine Learning
(UG Program: B.E. in Artificial Intelligence and Machine Learning)

By

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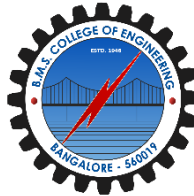
Semester & Section: 8 B

Under the Guidance of
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Assistant Professor

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Bengaluru – 19



DEPARTMENT OF MACHINE LEARNING

B.M.S. COLLEGE OF ENGINEERING

(An Autonomous Institute, Affiliated to VTU)

P.O. Box No. 1908, Bull Temple Road, Bengaluru - 560 019

May - 2026

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CERTIFICATE

This is to certify that Mr. **Pranav Veeraghanta** bearing USN: **IBM22AI092** , has satisfactorily completed the Course - Internship (Course code: 24AM8SRINT) with the title **“AI & Full Stack Development Intern”** in partial fulfillment of academic curriculum requirements of the 8th semester UG Program – B. E. in Artificial Intelligence and Machine Learning in the Department of Machine Learning, BMSCE, an Autonomous Institute, affiliated to Visvesvaraya Technological University, Belagavi during May 2026. It is also stated that the base work considered for the presentation is used only for academic purpose and not used in its original form anywhere for award of any degree

Student Signature

Signature of Guide

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Assistant Professor

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Signature of Head

Dr. Dakshayini M

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External Examination

Examiner Name and Signature

1.

2.

INTERNSHIP DETAILS

Name of Company/Organization:	CloudMpower Technologies Private Limited
Name of the Chairman/Head:	Ravishakar Subrahmanyam
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Website:	Cloudmpower.ai
Starting date of internship:	05/01/2026
Completion date of internship:	15/06/2026

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I would like to thank all the **Staff**, Department of Machine Learning for their support and encouragement during this assignment.

Special thanks to “**CloudMpower Technologies Private Limited**” who accepted me as an intern and gave me the opportunity to work and learn with them. I with deep sense of gratitude give sincere thanks to “**Mr Ravishankar Subrahmanyam, Founder and CEO of CloudMpower Technologies Private Limited**”

Last but not the least, I would like to thank my parents, family, and friends, without whose help and encouragement this work would have been impossible.

Pranav Veeraghanta
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EXECUTIVE SUMMARY

During my internship at CloudMPower Technologies Private Limited, I worked as an AI & Full Stack Development Intern, contributing to the development of intelligent web platforms, edge AI systems, and enterprise-focused AI workflows. My work involved building and optimizing real-time detection systems, AI-assisted reporting and dashboard platforms, conversational AI systems, semantic search workflows, and audio generation pipelines. I contributed across frontend development, backend integrations, UI/UX redesign, performance optimization, and cross-platform progressive web application support for web, Android, and iOS environments. The internship also involved extensive experimentation with browser-based AI runtimes, NLP systems, event-driven architectures, and intelligent workflow design, with a strong focus on making advanced AI systems more intuitive, practical, and user-friendly for real-world applications.

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Chapter 1

INTRODUCTION

1.1 Organization Overview

CloudMPower Technologies Private Limited is a Bengaluru-based technology company focused on developing AI-driven and enterprise software solutions. The organization works on applications involving artificial intelligence, multimodal systems, intelligent workflows, and enterprise automation.

The company emphasizes practical and production-ready implementations of AI technologies across web and enterprise environments.

1.2 Objectives of the Internship

The primary objective of the internship was to gain practical exposure to AI-powered product development and enterprise software engineering. The internship focused on understanding how intelligent systems are designed, optimized, and deployed for real-world usage.

The work involved contributing to products such as Detect, Insight, and AI Delve while improving technical skills in frontend and backend development, AI integration, UI/UX design, performance optimization, and cross-platform application development.

1.3 Scope of Work

The scope of work included developing AI-assisted web platforms, real-time intelligent systems, dashboard and reporting solutions, and progressive web applications.

The internship involved working on Detect, an edge AI-based detection system; Insight, an enterprise reporting and operations platform; and AI Delve, an AI-powered research and workflow platform. The work also covered computer vision systems,

NLP-based workflows, semantic search, conversational systems, audio generation pipelines, API integration, cross-platform optimization, performance testing, and deployment-related improvements.

Chapter 2

DESCRIPTION OF PROJECTS / TASKS

2.1 Tasks Assigned

The internship involved tasks related to AI system development, frontend and backend integration, workflow optimization, and enterprise application design.

Responsibilities included building and optimizing the Detect edge AI system, developing reporting and dashboard workflows for Insight, and implementing AI-assisted conversational and research workflows in AI Delve.

Additional tasks included improving UI/UX, integrating APIs, optimizing runtime performance, implementing progressive web application support, and ensuring cross-platform compatibility.

2.2 Plan of Action

The internship work began with research and experimentation involving AI models, browser runtimes, edge inference systems, and enterprise workflow structures. Prototype systems were developed and validated before integration into production-ready applications.

Detect focused on experimentation with object detection models, browser-based runtimes, and lightweight AI pipelines. Insight focused on reporting workflows, synchronization systems, dashboard functionality, and operational reliability.

AI Delve focused on AI-assisted workflows, semantic search, conversational systems, audio generation, and enterprise-oriented UI/UX design. Continuous improvements were carried out through debugging, testing, performance optimization, and iterative redesign.

2.3 Accomplishments

Successfully contributed to the development of multiple AI-powered systems during the internship. Detect evolved into a real-time edge AI detection platform supporting

lightweight and transformer-based workflows with offline functionality and runtime optimization. Insight was expanded into a full-stack reporting and operational management platform with dashboard systems, synchronization handling, GPS-aware workflows, and mobile-compatible progressive web application support.

AI Delve developed into an AI-powered research and workflow platform with semantic search, conversational systems, audio generation pipelines, notebook-based workflows, report-building systems, and enterprise-focused UI/UX redesigns. Multiple production-ready features and workflows were successfully implemented and deployed across these systems.

Chapter 3

TECHNICAL DETAILS

3.1 System Design and Architecture

The developed systems followed modular architectures consisting of frontend interfaces, backend APIs, AI inference pipelines, and real-time communication systems.

Detect utilized browser-based AI runtimes, lightweight inference frameworks, and progressive web application architecture to support edge-compatible workflows and offline functionality. Insight was structured around client systems, dashboards, reports, synchronization mechanisms, and administrative workflows.

AI Delve was designed around workspaces, notebooks, reports, chats, semantic search systems, and modular AI interaction flows using concepts such as lenses, modes, and blocks. Event-driven communication and responsive design principles were implemented across all systems.

3.2 Technologies and Implementation

The implementation involved frontend frameworks, browser-based AI inference systems, API integrations, server-sent events, semantic search workflows, and audio generation pipelines.

Detect involved experimentation with lightweight and transformer-based object detection models, runtime optimization, and segmentation workflows. Insight involved implementation of reporting systems, dashboard interfaces, synchronization handling, GPS-aware validation systems, and progressive web application support for Android and iOS environments.

AI Delve involved implementation of conversational AI systems, semantic search, report generation workflows, audio pipelines using language and text-to-speech models,

multi-workspace systems, and enterprise-oriented UI/UX redesigns. Extensive testing and optimization were carried out to improve stability, responsiveness, and cross-platform compatibility.

Chapter 4

SKILLS DEVELOPED

4.1 Technical Skills

- AI-powered web application development
- Edge AI and browser-based inference systems
- Computer vision workflows and runtime optimization
- Semantic search and NLP workflow implementation
- Conversational AI and audio generation systems
- Frontend and backend integration
- Progressive web application development
- Real-time communication systems using SSE
- Cross-platform optimization for web, Android, and iOS
- UI/UX design and responsive interface development
- API integration, debugging, and system validation
- Enterprise workflow and dashboard development

4.2 Soft Skills and Workplace Skills

- Problem-solving and analytical thinking
- Research-driven experimentation and learning
- Team collaboration and technical communication
- Adaptability to changing requirements and workflows
- Understanding of enterprise usability and user-centric design
- Time management and iterative development practices
- Documentation, reporting, and presentation skills
- Exposure to stakeholder discussions and product-oriented thinking
- Ability to balance technical implementation with user experience considerations

Chapter 5

Results

The internship provided practical exposure to the complete lifecycle of AI-powered software development, from research and experimentation to deployment and optimization. Contributions were made toward Detect, an edge AI-based detection platform; Insight, an enterprise reporting and operational workflow system; and AI Delve, an AI-powered research and workflow platform.

The implemented systems improved workflow efficiency, usability, responsiveness, and accessibility across web and mobile environments. The internship strengthened understanding of applied AI, software engineering, enterprise-focused UI/UX design, intelligent workflows, and real-world product development practices while contributing to multiple production-ready systems.

Chapter 6

LEARNING OUTCOMES

- Gained practical exposure to the complete lifecycle of AI-powered software development, including research, implementation, optimization, testing, and deployment.
- Developed strong understanding of frontend and backend integration, progressive web applications, semantic search systems, and real-time communication workflows.
- Improved ability to design user-centric AI systems by combining technical implementation with UI/UX principles and workflow optimization.
- Strengthened problem-solving, debugging, performance optimization, and cross-platform development skills through real-world product development.
- Learned how enterprise software systems are structured, validated, and improved through iterative testing, stakeholder feedback, and continuous refinement.

Chapter 7

Sustainable Development Goals (SDGs) Addressed

SDG 8 – Decent Work and Economic Growth

The internship involved developing enterprise-oriented platforms that improve workflow efficiency, reporting systems, intelligent research assistance, and operational management. These systems help streamline digital processes, support productivity, and contribute toward more efficient technology-driven workplaces and business environments.

SDG 9 – Industry, Innovation and Infrastructure

The internship contributed toward building scalable and intelligent digital systems using AI-powered workflows, real-time communication systems, and enterprise software platforms. The work focused on improving accessibility, efficiency, and usability of modern AI-driven applications across multiple platforms and devices.

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Chapter 8

Conclusion

The internship at CloudMpower Technologies Private Limited has proved valuable in many ways. As a problem solver, I have gained a platform where I am able to work on real world problems using real time systems and solutions. As engineers and developers, our worldview is often highly technical. Through decades of technological evolution, the focus has primarily been on building systems around functionality. However, as automation and intelligent systems continue to grow, it becomes increasingly important to focus on user experience and leverage inference and ML models to create meaningful and worthwhile experiences.

It is not about fitting the person to the system, it is about fitting the system to the person. Regardless of the domain a product belongs to, many systems still expose their complexity directly to users, making even small tasks feel unnecessarily heavy. The company has opened my eyes to thinking from this perspective, for which I am truly grateful. As I continue to learn, build, and explore, I seek to gain more knowledge by welcoming challenges and facing them instead of avoiding them.